

FAIRFIELD, IL, USA

WMO: 744656

Lat: 38.379N

Lon: 88.413W

Elev: 436

StdP: 14.47

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.6	12.1	-3.2	4.7	12.7	1.2	5.9	16.3	20.7	49.4	18.8	47.3	3.4	300	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	93.4	75.3	91.2	75.4	89.9	75.1	80.7	88.3	79.1	87.0	77.7	85.7	6.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	152.7	85.7	76.9	142.3	83.9	75.1	134.0	82.2	44.7	88.4	42.8	87.7	41.4	85.1	91.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	16.2	13.2	DB	0.1	97.0	4.7	3.2	-3.3	99.3	-6.0	101.2	-8.6	103.0	-12.0	105.4	
			WB	-0.7	83.3	4.5	2.9	-4.0	85.4	-6.6	87.0	-9.1	88.7	-12.3	90.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.4	31.9	35.1	46.4	57.1	66.6	75.2	77.7	75.9	70.1	57.6	45.7	36.7
	DBStd	18.43	11.54	11.34	10.98	9.26	8.26	5.72	4.93	5.10	6.69	9.36	9.82	10.01
	HDD50	1898	570	433	204	39	1	0	0	0	0	29	195	427
	HDD65	4566	1025	838	584	265	86	6	0	1	28	272	582	879
	CDD50	4242	9	14	91	251	516	755	858	802	603	264	66	13
	CDD65	1435	0	0	5	27	136	310	393	339	181	42	2	0
	CDH74	13598	0	0	31	215	1143	3064	3939	3151	1660	382	13	0
	CDH80	5345	0	0	1	23	335	1246	1702	1302	629	106	1	0
Wind	WSAvg	5.0	6.2	6.2	6.7	6.6	5.2	4.2	3.5	3.1	3.5	4.6	5.4	5.5
Precipitation	PrecAvg	44.2	3.6	2.5	3.6	4.4	5.0	4.0	3.6	3.3	3.0	3.5	4.1	3.2
	PrecMax	55.4	8.7	4.9	9.0	10.5	10.8	9.1	6.9	8.8	8.0	8.6	9.1	7.0
	PrecMin	36.5	0.6	0.4	0.9	1.3	1.3	1.1	1.0	0.9	0.1	0.6	0.1	1.7
	PrecStd	6.1	1.9	1.3	1.9	2.4	2.7	2.1	1.7	2.0	1.9	2.1	2.1	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	68.9	78.9	82.3	90.4	96.7	99.3	97.4	93.1	89.1	75.1	64.9
		MCWB	60.4	60.5	63.6	66.0	73.6	73.9	75.8	74.4	70.4	66.6	61.5	59.7
	2%	DB	60.9	63.6	72.6	80.0	86.4	92.3	93.2	93.2	90.4	82.5	70.2	61.0
		MCWB	56.1	56.6	62.1	64.3	71.5	75.2	78.0	74.8	71.8	67.6	61.2	57.2
	5%	DB	54.9	57.9	68.3	76.0	83.9	90.3	90.9	90.4	87.6	78.8	65.6	55.9
		MCWB	51.1	52.4	59.5	63.1	70.1	74.9	77.2	75.7	72.1	64.5	58.2	51.9
	10%	DB	48.4	53.0	63.7	72.6	81.2	87.9	88.7	87.7	83.5	73.2	61.5	52.1
		MCWB	44.6	47.5	56.9	61.3	69.3	73.3	76.4	75.6	69.9	62.9	54.1	48.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.0	62.3	67.3	70.4	77.0	81.2	83.5	83.4	78.1	73.9	67.0	61.9
		MCDB	64.4	66.4	73.7	78.8	85.2	88.9	92.8	91.1	84.8	84.0	70.7	63.9
	2%	WB	57.2	58.4	64.1	67.9	74.6	79.2	81.1	80.3	76.0	71.1	63.3	57.9
		MCDB	60.5	62.6	69.8	76.4	82.5	87.9	89.1	87.0	85.1	79.0	66.4	61.0
	5%	WB	51.9	53.5	61.1	65.6	72.8	77.5	79.5	78.3	74.3	68.4	59.5	53.2
		MCDB	54.2	57.6	67.6	72.6	80.6	85.9	87.7	86.0	82.8	74.8	64.6	55.5
	10%	WB	44.8	47.3	57.8	63.1	71.0	75.7	78.0	76.7	72.6	65.2	55.2	48.1
		MCDB	48.0	52.2	62.9	69.6	78.5	83.8	85.8	84.3	79.7	71.0	60.1	50.8
Mean Daily Temperature Range	5% DB	MDBR	15.1	16.0	17.9	19.6	18.6	19.6	18.6	19.9	21.7	20.8	17.9	14.3
		MCDBR	20.8	23.5	22.3	23.1	21.5	22.0	21.0	23.6	25.1	25.2	23.4	18.6
		MCWBR	18.1	18.6	14.4	12.7	9.7	9.0	9.2	10.3	10.1	11.8	14.7	16.1
	5% WB	MCDBR	19.1	21.8	20.4	18.9	18.9	18.7	18.2	19.8	20.4	20.8	19.8	17.1
		MCWBR	18.3	19.2	13.9	12.3	10.1	9.5	9.8	10.5	10.1	11.9	16.2	16.3
Clear-Sky Solar Irradiance	taub		0.311	0.325	0.363	0.402	0.436	0.440	0.466	0.468	0.411	0.360	0.322	0.309
	taud		2.455	2.400	2.358	2.253	2.213	2.221	2.156	2.140	2.291	2.430	2.480	2.470
	Ebn at Noon		275	286	284	278	269	267	259	255	265	270	269	268
	Edh at Noon		26	31	36	43	45	45	47	47	38	30	25	24
All-Sky Solar Radiation	RadAvg		626	869	1184	1562	1757	2031	1968	1787	1489	1083	730	534
	RadStd		61	94	111	90	152	131	125	97	102	113	82	50

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140

Nomenclature: See separate page